

# Markdown

Lightweight prose, but ket, bra, and integrals remain embedded TeX; complex structures and numbering stay manual.

## # Quantum States and Measurement

Consider a two-level system:

- $|\!0\rangle, |\!1\rangle$  form an orthonormal basis;
- $|\psi\rangle = \frac{1}{\sqrt{2}}(|\!0\rangle + i|\!1\rangle)$ .

### ## Definition 1.1 (State expansion)

```

$$
|\psi(t)\rangle = \sum_n c_n(t) |n\rangle
$$
i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle
= \hat{H} |\psi(t)\rangle
$$
The expectation value of  $\hat{A}$  is
$$
\langle \psi | \hat{A} | \psi \rangle
= \int \psi^*(x) \hat{A} \psi(x) dx
$$

```

### ## Theorem 1.1 (Normalization)

If  $\hat{H} = \hat{H}^\dagger$ , then  $\sum_n |c_n(t)|^2 = 1$  for every  $t$ .

- > **Proof.** Differentiate Definition 1.1
- > and use the Hermiticity of the Hamiltonian.

### ## Problem 1.1

Measure

$|\psi\rangle = \frac{1}{\sqrt{2}}(|\!0\rangle + i|\!1\rangle)$ .

Find the probabilities of the two basis states.

**Solution.**

$P(0) = |\langle 0 | \psi \rangle|^2 = \frac{1}{2}$ ,  
 $P(1) = |\langle 1 | \psi \rangle|^2 = \frac{1}{2}$ .

| State          | Amplitude            | Probability   |
|----------------|----------------------|---------------|
| $ \!0\rangle$  | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ |
| $i \!1\rangle$ | $\frac{i}{\sqrt{2}}$ | $\frac{1}{2}$ |

**Table 1.1 Measurement outcomes**

## Note Manual maintenance

Headings, itemize, and basic tables are direct; block styles, three-line tables, numbering, resets, and cross-references must be kept consistent by hand. Rendering options are limited too.

# LaTeX

Fully capable, but itemize, theorem blocks, ket/bra, and three-line tables require explicit commands or packages.

```
\usepackage{amsmath,amsthm,braket,booktabs}

\section{Quantum States and Measurement}
Consider a two-level system:
\begin{itemize}
\item  $\ket{0}, \ket{1}$  form an orthonormal
basis;
\item  $\ket{\psi}
=\frac{\ket{0}+i\ket{1}}{\sqrt{2}}$ .
\end{itemize}

\begin{definition}[State
expansion]\label{def:state}
\begin{aligned}
&\ket{\psi(t)}=\sum_n c_n(t)\ket{n} \\
&i\hbar\frac{\partial}{\partial t}\ket{\psi(t)} \\
&=\hat{H}\ket{\psi(t)}
\end{aligned}
The expectation value of  $\hat{A}$  is
\begin{aligned}
&\bra{\psi}\hat{A}\ket{\psi} \\
&=\int \psi^*(x)\hat{A}\psi(x)\,dx
\end{aligned}
\end{definition}
```

```
\begin{theorem}
[Normalization]\label{thm:norm}
If  $\hat{H}=\hat{H}^\dagger$ , then
 $\sum_n |c_n(t)|^2=1$  for every  $t$ .
\end{theorem}

\begin{proof}
Differentiate Definition~\ref{def:state}
and use the Hermiticity of the Hamiltonian.
\end{proof}

\begin{problem}
Measure
 $\ket{\psi}=\frac{\ket{0}+i\ket{1}}{\sqrt{2}}$ .
Find the probabilities of the two basis
states.
\end{problem}

\begin{solution}
 $P(0)=|\braket{0|\psi}|^2=\frac{1}{2}$ ,
 $P(1)=|\braket{1|\psi}|^2=\frac{1}{2}$ .
\end{solution}

\begin{table}
\centering
\begin{tabular}{ccc}
\toprule
State & Amplitude & Probability \\
\midrule
 $\ket{0}$  &  $\braket{0|\psi}=1/\sqrt{2}$  &  $1/2$  \\
 $\ket{1}$  &  $\braket{1|\psi}=i/\sqrt{2}$  &  $1/2$  \\
\bottomrule
\end{tabular}
\caption{Measurement outcomes}
\end{table}
```

## Note Boilerplate

The semantics are precise and the controls are deep, but you also maintain environments, labels, references, packages, and the build toolchain. Backslash-heavy source is harder to scan, and large local builds can interrupt the writing flow.

# Typst + Scripst

Lists, physics notation, and formulas stay compact; countblocks and three-line tables use ready-made styles.

## Quantum States and Measurement

Consider a two-level system:

- $|\psi_0\rangle, |\psi_1\rangle$  form an orthonormal basis;
- $|\psi\rangle = (|\psi_0\rangle + i |\psi_1\rangle)/\sqrt{2}$ .

```
#definition(subname: [State expansion], lab:
"def:state")[
$
ket(Psi(t)) = sum_n c_n (t) ket(n)
$
$
i hbar dv(, t) ket(Psi(t)) = hat(H)
ket(Psi(t))
$
The expectation value of  $\hat{A}$  is
$
braket(Psi, hat(A), Psi)
= integral Psi^*(x) hat(A) Psi(x) dif x
$
]
```

```
#theorem(subname: [Normalization], lab:
"thm:norm")[
If  $\hat{H} = \hat{H}^\dagger$ , then
 $\sum_n |\langle \psi_n | \psi(t) \rangle|^2 = 1$  for every  $t$ .
]

#proof[
Differentiate @def:state and use the
Hermiticity of the Hamiltonian.
]

#problem(lab: "prob:measure")[
Measure  $|\psi\rangle = (|\psi_0\rangle + i |\psi_1\rangle)/\sqrt{2}$ .
Find the probabilities of the two basis
states.
]

#solution[
 $P(0) = |\langle \psi_0 | \psi \rangle|^2 = 1/2$ ,
 $P(1) = |\langle \psi_1 | \psi \rangle|^2 = 1/2$ .
]

#figure(
three-line-table[
| State | Amplitude | Probability |
|  $|\psi_0\rangle$  |  $\langle \psi_0 | \psi \rangle = 1/\sqrt{2}$ 
|  $1/2$  |
|  $|\psi_1\rangle$  |  $\langle \psi_1 | \psi \rangle = i/\sqrt{2}$ 
|  $1/2$  |
],
caption: [Measurement outcomes],
)
```

## Note Scripst + Ratchet

Scripst integrates Physica, Tablem, and content-block presets, while Ratchet keeps numbering depth, resets, and references consistent.

# Real Script Output

These are not imitation boxes: the physics notation, three-line table, and content blocks below are generated by Script itself.

## Quantum States and Measurement

Consider a two-level system:

- $|0\rangle, |1\rangle$  form an orthonormal basis;
- $|\Psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$ .

### Definition 1.1 State expansion

$$|\Psi(t)\rangle = \sum_n c_n(t) |n\rangle \quad (1.1)$$

and satisfies

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad (1.2)$$

The expectation value of  $\hat{A}$  is

$$\langle \Psi | \hat{A} | \Psi \rangle = \int \Psi^*(x) \hat{A} \Psi(x) dx \quad (1.3)$$

### Theorem 1.1 Normalization

If  $\hat{H} = \hat{H}^\dagger$ , then  $\sum_n |c_n(t)|^2 = 1$  for every  $t$ .

*Proof.* Differentiate [Definition 1.1](#) and use the Hermiticity of the Hamiltonian.

■

### Problem 1.1

Measure  $|\Psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$  and find the probabilities of the two basis states.

*Solution.*  $P(0) = |\langle 0 | \Psi \rangle|^2 = \frac{1}{2}$ , and  $P(1) = |\langle 1 | \Psi \rangle|^2 = \frac{1}{2}$ .

| State       | Amplitude                                       | Probability   |
|-------------|-------------------------------------------------|---------------|
| $ 0\rangle$ | $\langle 0   \Psi \rangle = \frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ |
| $ 1\rangle$ | $\langle 1   \Psi \rangle = \frac{i}{\sqrt{2}}$ | $\frac{1}{2}$ |

Table 1.1: Measurement outcomes